Statistical Analysis for Data Science — Titanic

Solutions for all tasks in the statistics assignment. The source file specifies the Titanic dataset. The PDF provides executable analysis code; numerical test results are generated when run against the selected Titanic dataset rather than fabricated here.

Task 1. Load and explore Titanic dataset

The assignment explicitly says to use the Titanic dataset. Download the Kaggle train.csv file and place it beside the notebook, or use the seaborn dataset option. The code below uses seaborn's Titanic dataset.

Python / SQL code

```
import pandas as pd
import seaborn as sns

titanic = sns.load_dataset("titanic")
print(titanic.head())
print(titanic.shape)
print(titanic.info())
print(titanic.describe(include="all"))
```

Sample output

```
The first rows, dimensions, data types and descriptive summary are printed.
The exact output depends on the version/source of the Titanic data.
```

Task 2. Summary statistics, skewness and kurtosis

Compute these measures for numeric columns. Pandas variance/std are sample statistics; skew and kurtosis are sample estimates.

Python / SQL code

```
numeric = titanic.select_dtypes(include="number")
summary = pd.DataFrame({
    "mean": numeric.mean(),
    "median": numeric.median(),
    "mode": numeric.mode().iloc[0],
    "variance": numeric.var(),
    "std_dev": numeric.std(),
    "skewness": numeric.skew(),
    "kurtosis": numeric.kurt()
})
print(summary)
```

Sample output

```
A statistics table is printed for numeric Titanic features such as
survived, pclass, age, sibsp, parch and fare (depending on dataset version).
Missing values are skipped by Pandas calculations.
```

Task 3. Distributions, boxplots, Q-Q plot and Shapiro-Wilk

Use a numeric feature such as fare or age. Shapiro-Wilk is sensitive to large sample sizes; interpret its p-value with plots and context.

Python / SQL code

```
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

feature = titanic["age"].dropna()
sns.histplot(feature, kde=True)
plt.title("Titanic passenger age distribution")
plt.show()

sns.boxplot(x=feature)
plt.title("Age boxplot")
plt.show()

stats.probplot(feature, dist="norm", plot=plt)
plt.title("Q-Q plot for age")
plt.show()

sample = feature.sample(n=min(5000, len(feature)), random_state=42)
statistic, p_value = stats.shapiro(sample)
print("Shapiro statistic:", statistic)
print("p-value:", p_value)
print("Reject normality at 0.05:", p_value < 0.05)
```

Sample output

```
Three plots are displayed. The Shapiro-Wilk statistic and p-value are
printed from the selected Titanic data. If p < 0.05, reject the null
hypothesis that the sample comes from a normal distribution.
```

Task 4A. Independent t-test: compare average fares by survival

Example hypotheses: H0: mean fare is equal for survivors and non-survivors. H1: the means differ. Welch's t-test does not assume equal variances.

Python / SQL code

```
from scipy.stats import ttest_ind

survived_fare = titanic.loc[titanic["survived"] == 1, "fare"].dropna()
not_survived_fare = titanic.loc[titanic["survived"] == 0, "fare"].dropna()
t_stat, p_value = ttest_ind(survived_fare, not_survived_fare,
                            equal_var=False)
print("H0: mean fares are equal")
print("t statistic:", t_stat)
print("p-value:", p_value)
print("Significant:", p_value < 0.05)
```

Sample output

```
The program prints the t statistic and p-value.
If p < 0.05, there is evidence of a difference in average fare
between the two survival groups. This is an association, not proof
that fare caused survival.
```

Task 4B. Chi-square test: sex and survival

H0: sex and survival are independent. H1: they are associated.

Python / SQL code

```
from scipy.stats import chi2_contingency

table = pd.crosstab(titanic["sex"], titanic["survived"])
chi2, p_value, dof, expected = chi2_contingency(table)
print(table)
print("Chi-square:", chi2)
print("Degrees of freedom:", dof)
print("p-value:", p_value)
print("Significant:", p_value < 0.05)
```

Sample output

```
Contingency table and test statistics are printed.
If p < 0.05, reject independence: sex and survival are statistically
associated in this dataset.
```

Task 5A. Correlation matrix and heatmap

Correlations describe linear association between numeric columns and do not establish causation.

Python / SQL code

```
corr = numeric.corr()
print(corr)
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Titanic numeric-feature correlation")
plt.show()
```

Sample output

```
Prints the correlation matrix and displays a heatmap.
Values close to +1 indicate positive linear association; values
close to -1 indicate negative linear association.
```

Task 5B. Linear regression: predict fare

Use a simple, interpretable model predicting fare from passenger class, age, and family counts. Missing rows are removed for this demonstration.

Python / SQL code

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

model_data = titanic[["fare", "pclass", "age", "sibsp", "parch"]].dropna()
X = model_data[["pclass", "age", "sibsp", "parch"]]
y = model_data["fare"]
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
print("R-squared:", r2_score(y_test, predictions))
print("RMSE:", mean_squared_error(y_test, predictions) ** 0.5)
```

Sample output

```
The model prints its intercept, coefficients, R-squared and RMSE.
R-squared measures the proportion of test-set variance explained;
RMSE is the typical prediction error in fare units.
```

Task 7. Final analysis and report

Use the actual printed statistics and plots from the run to complete the report. Do not invent numerical findings before executing on the chosen dataset.

Python / SQL code

```
# Example report template
# 1. Descriptive statistics:
#    Report mean, median, spread, skewness and kurtosis for key variables.
# 2. Distribution:
#    Describe histogram/boxplot shape and Q-Q plot departures from normality.
#    Report Shapiro-Wilk p-value and alpha = 0.05 decision.
# 3. Hypothesis tests:
#    Report t-test and chi-square statistics, p-values and decisions.
# 4. Correlation/regression:
#    Describe strongest correlations; report R-squared and RMSE.
# 5. Business impact:
#    Explain how passenger class/fare/other associations could inform
#    customer segmentation or service planning; avoid causal claims.
```

Sample output

```
Conclusion example (to be finalized after running):
The analysis describes passenger demographics, ticket fares and survival.
Hypothesis tests assess differences in fare by survival and association
between sex and survival. Regression evaluates how well class, age and
family counts predict fare. All conclusions must quote the actual outputs.
```